

Geometry

Unit 8

Lesson 7 Homework

Name **Answer Key** _____

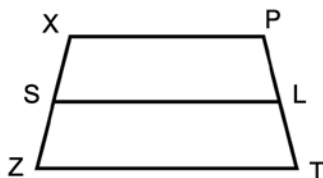
Date _____

Period _____

$XZTP$ is a trapezoid where $\overline{XP} \parallel \overline{ZT}$. $\overline{SL}$ is a midsegment of $XZTP$ and $XS = 7$, $LT = 6$, $SL = 10$, $TZ = 2x - 4$, and $XP = x - 3$.

- Write an equation to show the relationship between $\overline{SL}$ and the bases of the trapezoid.

$$SL = \frac{XP + ZT}{2}$$



- What is the value of x ?

$$\begin{aligned} 10 &= \frac{2x - 4 + x - 3}{2} \\ 20 &= 2x - 4 + x - 3 \\ 27 &= 3x \\ x &= 9 \end{aligned}$$

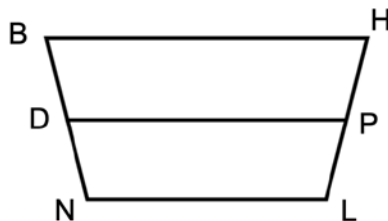
- What is the length of $\overline{XP}$ and $\overline{ZT}$?

$$\begin{aligned} ZT &= 14 \\ XP &= 6 \end{aligned}$$

Trapezoid $BHLN$ is shown where $\overline{BH} \parallel \overline{NL}$ with midsegment $\overline{DP}$ and $BH = 5x + 12$, $NL = 3x$, and $DP = 5x$.

- Find the value of x .

$$\begin{aligned} 5x &= \frac{5x + 12 + 3x}{2} \\ 10x &= 8x + 12 \\ 2x &= 12 \\ x &= 6 \end{aligned}$$



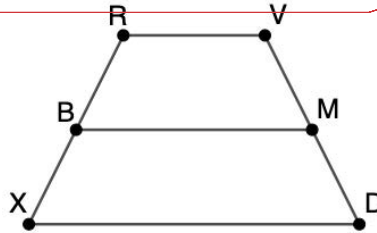
- Find the length of $\overline{DP}$.

$$DP = 30$$

- Find the lengths of $\overline{BH}$ and $\overline{NL}$.

$$\begin{aligned} BH &= 42 \\ NL &= 18 \end{aligned}$$

Isosceles trapezoid $RVDX$ is shown with midsegment $\overline{BM}$ where $RV = 2x + 1$, $XD = 3x + 2$, $BM = 2x + 4$ and $MD = 3$.



Commented [DB1]: <https://www.geogebra.org/calculator/vcppzgw3>

7. What is the value of x ?

$$\begin{aligned} 2x + 4 &= \frac{2x + 1 + 3x + 2}{2} \\ 4x + 8 &= 5x + 13 \\ x &= 5 \end{aligned}$$

8. What is the length of $\overline{BM}$?

$$BM = 14$$

9. What is the length of $\overline{XD}$ and $\overline{RV}$?

$$\begin{aligned} XD &= 17 \\ RV &= 11 \end{aligned}$$

10. What is the perimeter of $RVDZ$?

$$\begin{aligned} \text{Perimeter} &= RV + VD + XD + RX \\ \text{Perimeter} &= 11 + 2(3) + 17 + 2(3) \\ \text{Perimeter} &= 42 \end{aligned}$$